Lesson Plan: Soil vs. Lunar Regolith – Arduino Moisture Sensor Exploration

Standard (Next Generation Science Standards)

MS-ESS3-1 Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes.

Grade Level

Middle School

Time Required:

- Procedure: 1 class period (45-60 min)
- Discussion & Analysis: 1 class period (45-60 min)

Lesson Overview:

Students will test two materials and determine their moisture content using an Arduino Soil Moisture Sensor.

Learning Objectives:

Students will be able to:

1. Create a controlled experiment with two conditions
2. Use and record data from a soil moisture sensor
3. Connect findings to larger systems

Key Vocabulary:

- Soil
- Regolith (material on the moon's surface)
- Ecosystem
- Resource
- Variable (independent, dependent, controlled)

Materials:

- Topsoil or potting soil
- Lunar regolith simulant (or rock/clay/sand mixture)
- 12 oz plastic cups or other small pots
- Graduated cylinder
- Arduino soil moisture sensor
- Arduino UNO and SEEED sensor kit

Experimental Design

Independent Variable

Type of soil used:

- Topsoil/Potting soil
- Regolith simulant (or equivalent)

Dependent variable

- Moisture level (in Volts)

Controlled variables

- Amount of water added
- Amount of soil
- Container used for soil
- Sensor used

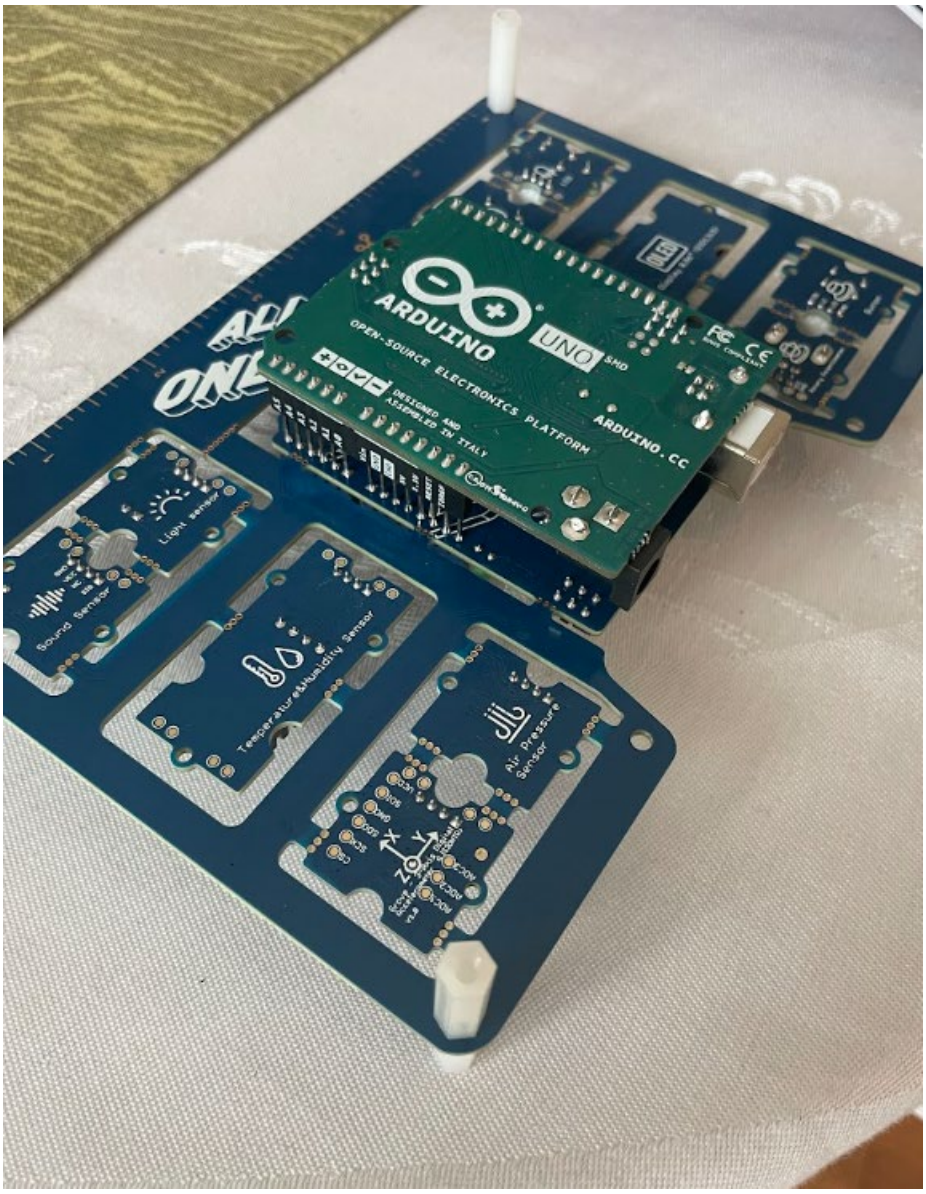
Setup



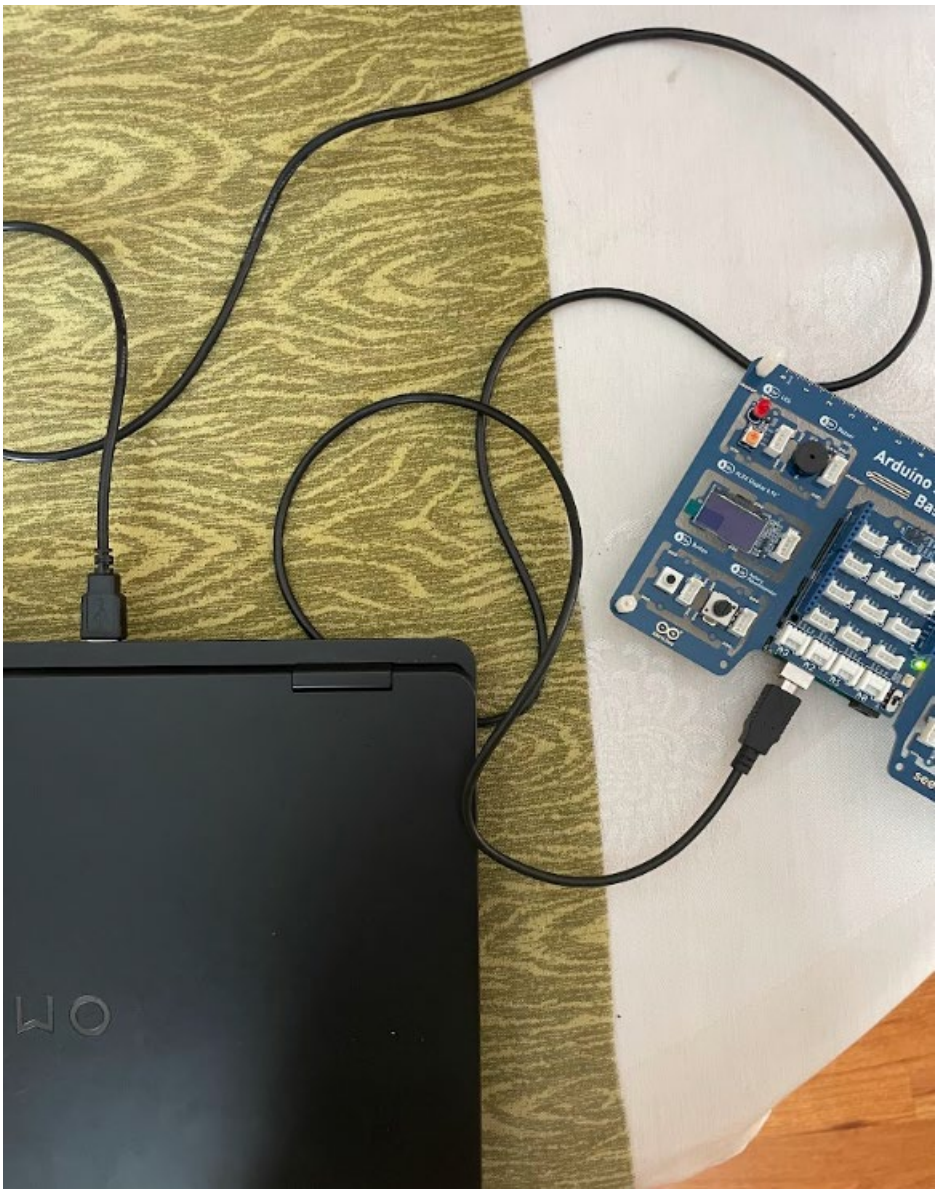
SEEED Arduino Sensor Kit
Source (2026 SEEED Technologies)

The Arduino, SEEED sensor kit, and moisture sensor can be used to monitor controlled variables such as soil moisture and light levels. Be sure to check out calibrating the moisture sensor to adjust the dry and wet values for more accurate measurements.

Follow pictures to set up equipment, then copy code into Arduino IDE to track values from the sensor.



1. Attach Arduino Uno to the bottom of the Sensor Kit Base. (Ensure that the Uno is pushed in all the way.)



2. Use cord to connect the Uno to a computer. Check that green light is on (as shown in the image) to ensure that the base is powered.

Add moisture sensor pictures here?

Code for moisture sensor:

```
#include "Arduino_SensorKit.h"
float moistureval;
float wetval=2.49;
float dryval=3.56;
```

```
void setup() {
Oled.begin();
Oled.setFlipMode(true); // Sets the rotation of the screen
}

void loop() {
Oled.setFont(u8x8_font_chroma48medium8_r);
Oled.setCursor(0, 33); // Set the Coordinates

// Reading the moisture sensor and converting the input to voltage
moistureval = analogRead(A1)*5.0/1023.0; //Scaling the moisture sensor based on the
calibration values to 100% moisture
moistureval = (moistureval-dryval)/(wetval-dryval) * 100.0;

//Printing the moisture to the OLED screen
Oled.print(" Moist: ");
Oled.print(moistureval);
Oled.println(" %");
Oled.refreshDisplay();
delay(1000); // runs once per second
}
```

Procedure

1. Introduce lunar regolith and challenges associated with growing plants on the Moon.
2. Students record an initial hypothesis and predict if there will be differences in moisture content.
3. Students separate into groups and gather necessary tools.
4. Students examine the two substances and record qualitative observations on their worksheets.
5. After observing the regolith, students can revise their hypothesis and explain how their observations altered their predictions.
6. Groups measure 1/2 cup of potting soil and regolith in separate cups and add labels.
7. Using a graduated cylinder, add 100 mL of water to both cups.
8. Insert soil moisture sensor into each cup and record moisture content.
9. Record group data on the board. As a class, calculate average and discuss whether results were consistent.
10. Transition to analysis and conclusion questions.

Guiding Questions:

- What effect does soil type have on the surrounding ecosystem?
- How are lunar regolith and potting soil different? How would their properties affect plant growth?
- What physical properties contribute to any differences in moisture levels?
- Based on our data, what challenges would astronauts face in growing plants on the moon?
- What do you predict would happen if the two substances were mixed?

Concluding Activity (Claim, Evidence, Reasoning & What If?)

On their worksheets, students complete a short Claim, Evidence, Reasoning paragraph to summarize their findings on the connection between soil type and moisture retention. Students can also complete the What If? section to expand their understanding beyond the scope of the activity. (What if... you mixed potting soil and regolith? You doubled the amount of water added? You were designing a farming system on the moon?)